Reliable CMOS PUF Cell for IoT Hardware Security Using Schmitt-Trigger-Assisted Readout

Nadeen Jaber, Lana Sayes, Julnar Assi, Sadeen Faqih
Department of Electrical and Computer Engineering, Birzeit University
Digital Integrated Circuits Project

***Abstract*—Physically unclonable functions (PUFs) provide a lightweight hardware-security primitive for generating device-specific responses from intrinsic CMOS process variations without storing permanent secret keys. This paper presents the design and simulation of a compact SRAM-based CMOS PUF architecture for IoT hardware security. The proposed architecture combines a conventional 6T SRAM cell with a Schmitt-trigger-assisted readout circuit to improve sensing reliability. By introducing hysteresis through positive feedback, the Schmitt Trigger enhances noise immunity and reduces the sensitivity of the readout stage to weak startup states and metastability. The complete architecture was implemented hierarchically in Electric VLSI and verified using NGSpice simulations. The Schmitt Trigger achieved full CMOS logic swing with an output of approximately 5.00 V for a low input and 0.043 V for a high input, while exhibiting a switching threshold near 2.7 V under a 5 V supply. The proposed 1-bit readout circuit also generated valid digital logic levels of approximately 0.043 V and 4.64 V for complementary SRAM states. The obtained results demonstrate that Schmitt-trigger-assisted sensing improves the robustness of SRAM-PUF response extraction and provides a practical solution for compact IoT-oriented hardware security applications.**

***Index Terms*—CMOS, SRAM PUF, physically unclonable function (PUF), Schmitt Trigger, readout circuit, Electric VLSI, NGSpice, IoT hardware security.**

I. Introduction

Hardware security has become a fundamental requirement in Internet-of-Things (IoT) devices, where embedded systems operate under stringent area, power, and cost constraints. Conventional security mechanisms typically rely on storing secret cryptographic keys in non-volatile memory. However, permanent key storage increases hardware overhead and exposes the device to invasive physical attacks such as probing, reverse engineering, and memory extraction. Physically unclonable functions (PUFs) provide an attractive alternative by generating unique device-specific responses from the intrinsic manufacturing variations of integrated circuits rather than storing secret information permanently.

Among various PUF architectures, SRAM-based PUFs have attracted considerable attention because they utilize the natural startup behavior of standard 6T SRAM cells that already exist in many digital systems. During power-up, unavoidable transistor mismatch causes each SRAM cell to settle into one of two stable logic states. This startup value can be used as a unique response bit for device authentication and cryptographic key generation. However, cells with small mismatch may exhibit unstable startup behavior due to thermal noise, supply voltage fluctuations, temperature variation, and metastability, reducing the reliability of the generated response. Consequently, the design of a robust sensing and readout circuit is essential for improving the stability and repeatability of SRAM-PUF responses.

The objective of this work is to design and evaluate a compact CMOS SRAM-based physically unclonable function (PUF) architecture incorporating a Schmitt-trigger-assisted readout circuit for improved sensing reliability. The proposed architecture was implemented hierarchically using Electric VLSI and verified through transistor-level NGSpice simulations. The complete system includes a CMOS inverter, a 6T SRAM PUF cell, a Schmitt Trigger sensing stage, a 1-bit readout circuit, a WL decoder, a Bit Detector, and a 4×4 SRAM PUF array. By introducing hysteresis into the sensing stage, the proposed readout improves the robustness of SRAM startup-state detection while maintaining a compact CMOS implementation suitable for low-cost IoT hardware.

The main contributions of this work are summarized as follows:

- A complete hierarchical CMOS SRAM-PUF architecture was designed and implemented using Electric VLSI.
- A Schmitt Trigger sensing circuit was developed to improve the robustness of SRAM startup-state detection through hysteresis and positive feedback.
- A 1-bit readout circuit integrating the 6T SRAM cell with the Schmitt Trigger was implemented and verified using transistor-level NGSpice simulations.
- A wordline decoder, Bit Detector, and a compact 4×4 SRAM PUF array were integrated to demonstrate the complete hierarchical PUF architecture.

The remainder of this paper is organized as follows. Section II reviews the background and related work. Section III presents the proposed CMOS PUF architecture. Section IV describes the design and implementation of each circuit block. Section V discusses the layout implementation. Section VI presents the simulation methodology and experimental results. Finally, Section VII concludes the paper and outlines possible future work.

II. Background and Related Work

Physically unclonable functions (PUFs) exploit intrinsic CMOS process variations to generate chip-unique responses. In SRAM-based PUFs, the mismatch between the two cross-coupled inverter branches determines the preferred power-up

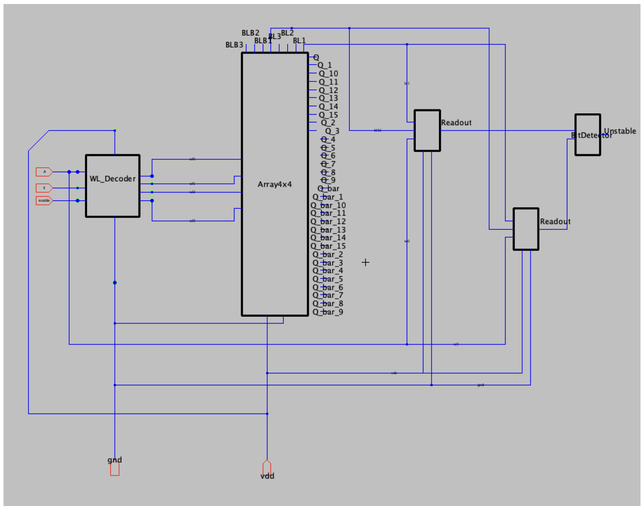

Fig. 1: High-level architecture of the proposed Schmitt-trigger-assisted SRAM PUF system. The 6T SRAM array generates the intrinsic PUF response, while the Schmitt Trigger enhances the stability of the sensed digital output.

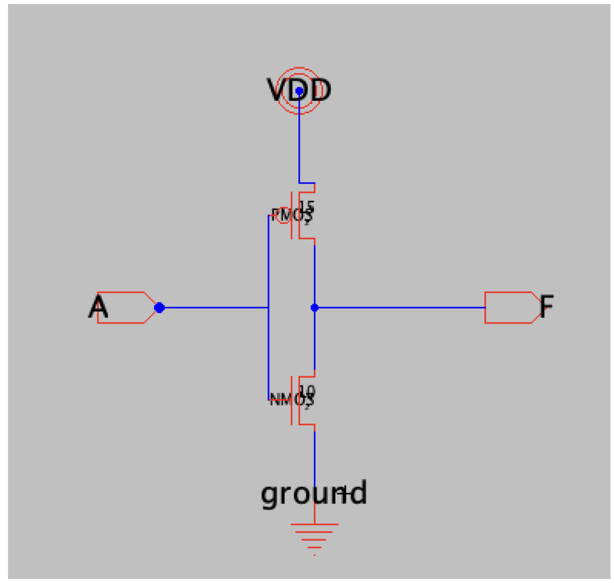

Fig. 2: Transistor-level schematic of the CMOS inverter implemented in Electric VLSI.

state of each cell. Ideally, each cell repeatedly generates the same response bit under different operating conditions. In practice, however, weak cells with small mismatch may behave unpredictably, which increases the bit error rate (BER) and reduces response reliability.

Several stabilization techniques have been proposed to improve PUF reliability, including error-correction codes (ECC), unstable-bit masking, temporal or spatial majority voting, and in-cell stabilization. The ASCH-PUF work emphasizes that conventional stabilization methods can introduce significant area, power, latency, or enrollment overhead, especially when targeting very low BER [1]. It also demonstrates that integrating detection and stabilization circuits close to the PUF array can reduce unstable-bit effects without relying only on expensive error-correction hardware.

This work does not attempt to reproduce the complete ASCH-PUF architecture. Instead, it adopts the main reliability motivation and applies it to a compact CMOS SRAM-PUF implementation by incorporating a Schmitt-trigger-assisted readout stage. The Schmitt Trigger is suitable for this purpose because its hysteresis improves immunity to voltage fluctuations around the switching point. Therefore, it can serve as a robust sensing interface between the SRAM PUF cell and the digital output stage.

III. Proposed Architecture

The proposed architecture is a compact 16-bit SRAM-based physically unclonable function (PUF) designed for lightweight IoT hardware security applications. As illustrated in Fig. 1, the system consists of a WL decoder, a 4×4 array of 6T SRAM PUF cells, a Schmitt-trigger-assisted readout circuit, and a Bit Detector stage. Each SRAM cell contributes one response bit, allowing the complete array to generate a unique 16-bit PUF response.

The operation of the proposed architecture is illustrated in Fig. 1. First, the WL decoder activates the target row of the array by driving the corresponding wordline while the remaining rows are held disabled. The complementary bitlines (BL and BLB) of the selected SRAM cell carry the startup state generated by transistor mismatch. The sensed voltage is then applied to the Schmitt Trigger, which converts the analog-like signal into a stable CMOS logic level by exploiting hysteresis and positive feedback. Finally, the generated response bit is passed to the Bit Detector, which evaluates the sensed output and forwards a valid digital response bit that can be combined with the remaining bits to form the complete 16-bit PUF response.

IV. Circuit-Level Design

A. CMOS Inverter

The CMOS inverter is the fundamental building block of the proposed SRAM-PUF architecture. It forms the cross-coupled storage element of the 6T SRAM cell and is also reused in several supporting digital blocks. The circuit consists of one PMOS transistor connected to the pull-up network and one NMOS transistor connected to the pull-down network. When the input voltage is low, the PMOS transistor conducts and charges the output node to V_{DD}, while the NMOS transistor remains off. Conversely, when the input voltage is high, the NMOS transistor turns on and discharges the output node to ground, producing the complementary logic level.

Figure 2 illustrates the transistor-level schematic of the CMOS inverter implemented in Electric VLSI.

The corresponding layout is shown in Fig. 3. The PMOS transistor is placed in the pull-up region, whereas the NMOS transistor is placed in the pull-down region following a conventional CMOS layout style. Dedicated V_{DD} and GND rails were routed to simplify hierarchical integration with larger circuit blocks such as the SRAM cell.

To verify the functionality of the inverter, a DC sweep simulation was performed using NGSpice. The input voltage

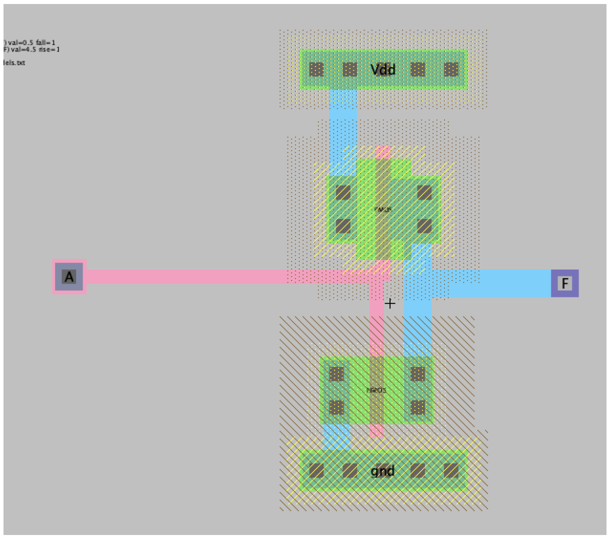

Fig. 3: Electric VLSI layout of the CMOS inverter showing the PMOS and NMOS transistors with dedicated power and ground rails.

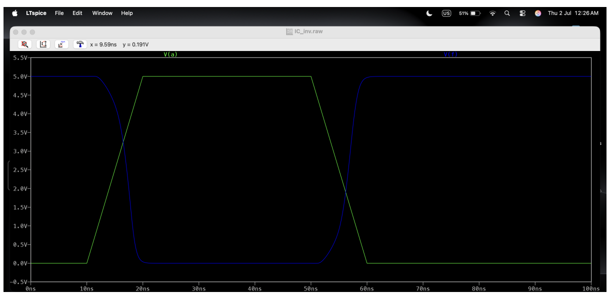

Fig. 4: NGSpice DC transfer characteristic of the CMOS inverter.

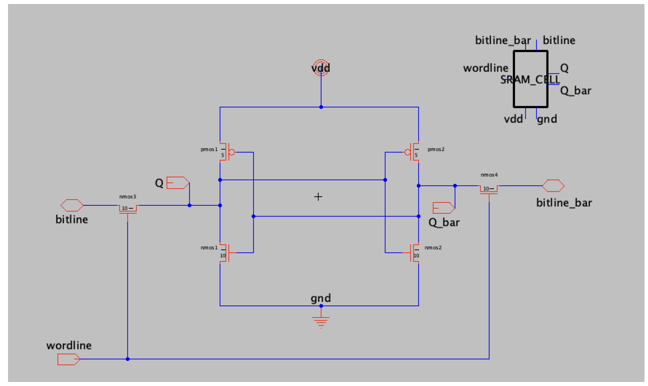

Fig. 5: Transistor-level schematic of the 6T SRAM PUF cell. The cross-coupled inverters generate the startup state, while the access transistors are controlled by WL.

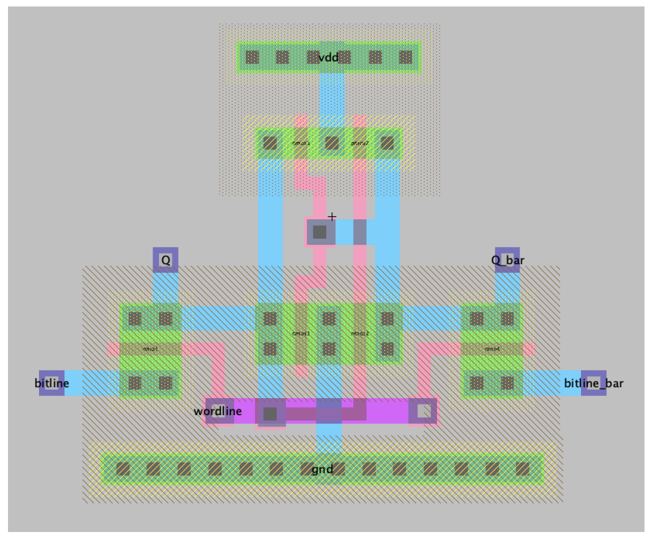

Fig. 6: Electric VLSI layout of the 6T SRAM PUF cell. Matching-aware placement is important because mismatch determines the PUF response.

was swept from 0 V to 5 V while monitoring the output voltage. As expected, the circuit produced a logic HIGH output for low input voltages and a logic LOW output for high input voltages, confirming correct inverter operation prior to its integration into the SRAM PUF cell.

The simulation results demonstrate the expected voltage-transfer behavior of the CMOS inverter, validating its functionality as the basic logic element used throughout the proposed PUF architecture. Since the inverter forms the cross-coupled storage pair inside the 6T SRAM cell, its correct switching behavior is a prerequisite for reliable PUF operation: any asymmetry between the two inverter branches is precisely the mismatch that the SRAM PUF exploits to generate a repeatable startup state. Confirming inverter functionality at this stage therefore provides a solid basis for the more complex blocks built on top of it.

B. 6T SRAM PUF Cell

The baseline PUF cell is based on a conventional 6T SRAM cell, consisting of two cross-coupled CMOS inverters and two access transistors controlled by the wordline. The complementary bitlines are denoted BL and BLB. During power-up, mismatch between the cross-coupled inverter branches causes the cell to settle into one of two stable states before the access transistors are even enabled. This startup state, rather than any externally written value, is what is exploited as the raw PUF response bit.

Fig. 5 shows the transistor-level SRAM cell schematic implemented in Electric VLSI. The two inverters are connected back-to-back so that the output of each drives the input of the other, forming a bistable latch, while the access transistors connect the internal storage nodes to BL and BLB whenever WL is asserted.

The SRAM layout, shown in Fig. 6, was designed with attention to compactness, device matching, and clean routing, since the physical symmetry (or intentional lack thereof) of the two inverter branches directly influences the mismatch that determines the cell's PUF response.

The cell was simulated in NGSpice to verify that it can store complementary internal states and can be correctly accessed through the wordline and bitline pair, as shown in Fig. 7.

The simulation confirms that the cell reliably holds one of

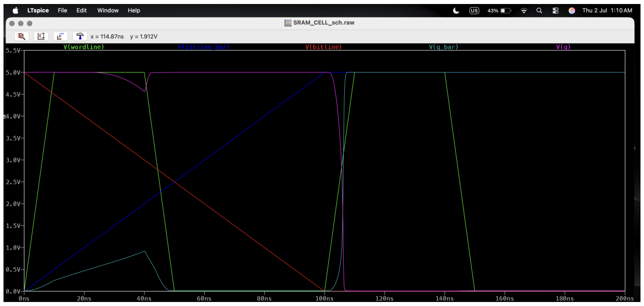

Fig. 7: NGSpice simulation of the SRAM PUF cell showing the complementary storage behavior.

TABLE I: Schmitt Trigger Transistor Sizing

Device	Width (μm)	Length (μm)
PMOS	0.72	0.36
NMOS	0.36	0.36

the two complementary states and that the stored node can be read out through the access transistors, which validates its use as the entropy source for the readout circuit described next.

C. Schmitt Trigger Sensing Circuit

The Schmitt Trigger is used as the sensing stage of the proposed PUF readout circuit. Since the voltage produced by an SRAM cell during startup or readout may be affected by noise, process variations, and metastability, a standard CMOS inverter cannot always be relied upon to provide a clean logic decision. The Schmitt Trigger addresses this by introducing hysteresis, which allows it to reject small voltage fluctuations near the switching point and improve the stability of the sensed logic value.

The implemented Schmitt Trigger is based on a CMOS inverter augmented with additional positive-feedback transistors. The circuit includes PMOS devices in the pull-up network, NMOS devices in the pull-down network, and feedback transistors controlled by the output node. This positive feedback shifts the effective switching threshold depending on whether the input voltage is rising or falling, producing two distinct trip points instead of one. The resulting hysteresis band reduces the effect of noise near the nominal switching point and prevents unwanted output oscillations that a conventional inverter would be susceptible to when driven by a slowly varying or noisy input, as is often the case with a weak SRAM startup state.

The transistor sizing used in the Schmitt Trigger design is summarized in Table I. The relative sizing of the feedback devices with respect to the main pull-up and pull-down transistors sets the width of the hysteresis band; larger feedback transistors widen the band and further improve noise rejection at the cost of a more asymmetric transfer characteristic.

Fig. 8 shows the Electric VLSI schematic of the proposed Schmitt Trigger, and the block was subsequently converted into a hierarchical icon to simplify its integration into the readout circuit.

The corresponding layout is shown in Fig. 9. The layout separates the pull-up and pull-down networks and keeps the

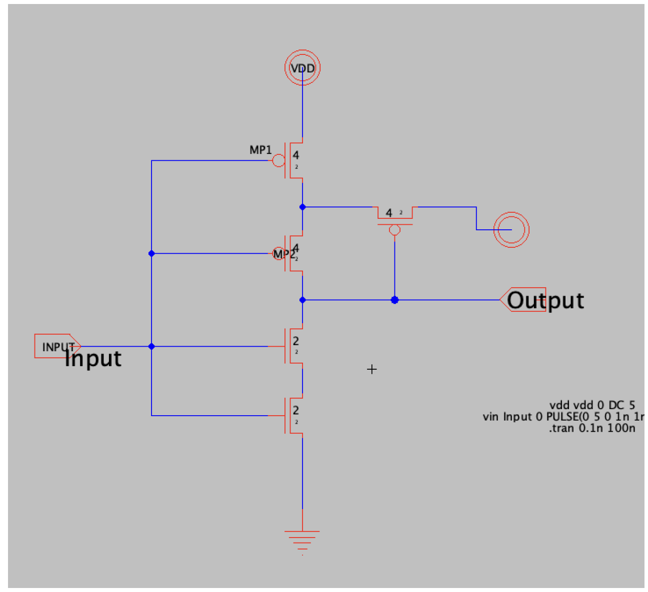

Fig. 8: Electric VLSI schematic of the Schmitt Trigger sensing circuit. Positive feedback improves noise immunity around the switching region.

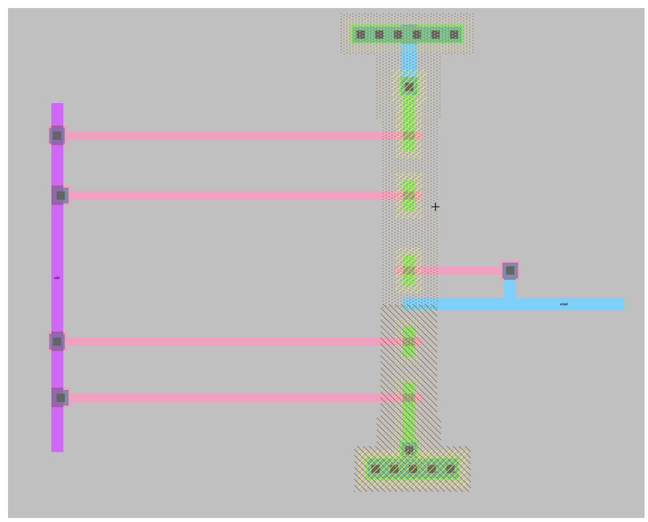

Fig. 9: Electric VLSI layout of the Schmitt Trigger sensing circuit.

feedback routing short and direct, which limits parasitic loading on the feedback path and helps preserve the intended hysteresis behavior.

To verify the design, a DC sweep simulation was performed in NGSpice with the supply voltage fixed at 5 V and the input voltage swept from 0 V to 5 V in steps of 0.05 V. Fig. 10 shows the resulting transfer characteristic.

The simulation confirmed correct digital operation and full logic swing. At $V_{in} = 0$ V, the output was approximately 5.00 V, and at $V_{in} = 5$ V, the output was approximately 0.043 V. The switching threshold occurred near

$$V_{TH} \approx 2.7 \text{ V}. \tag{1}$$

These results, summarized in Table II, confirm that the circuit

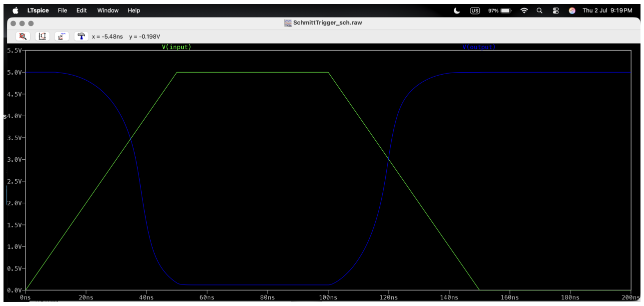

Fig. 10: NGSpice DC transfer characteristic of the proposed Schmitt Trigger.

TABLE II: Schmitt Trigger Simulation Results

Input	Output	Logic
0 V	≈ 5.00 V	HIGH
5 V	≈ 0.043 V	LOW

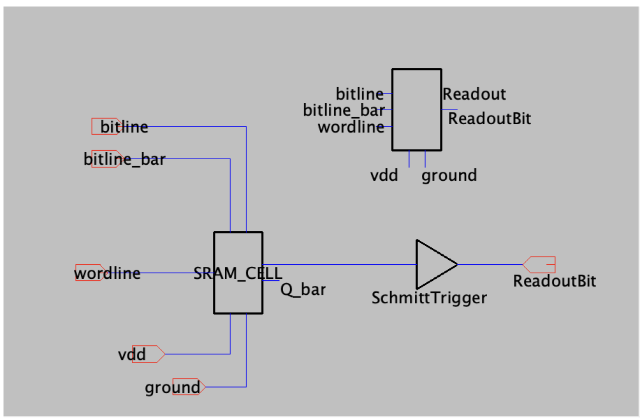

Fig. 11: Hierarchical schematic of the proposed 1-bit readout circuit combining the 6T SRAM PUF cell with the Schmitt Trigger sensing stage.

regenerates clean CMOS logic levels rather than passing through an intermediate or weakly defined voltage, which is the key property required for it to serve as a dependable sensing stage ahead of the readout circuit.

D. 1-Bit Readout Circuit

To improve the reliability of the generated PUF response, a dedicated 1-bit readout circuit was designed by directly connecting a 6T SRAM cell to a Schmitt Trigger sensing stage. In operation, the SRAM cell generates the startup state produced by transistor mismatch, and the Schmitt Trigger converts the sensed node voltage into a stable digital logic level. The readout block takes BL, BLB, WL, V_{DD}, and ground as inputs and produces a single output, ReadoutBit.

Fig. 11 shows the hierarchical schematic of the complete 1-bit readout block, and Fig. 12 shows the corresponding icon used to integrate the block into the array.

The layout of the 1-bit readout circuit combines the SRAM cell and Schmitt Trigger into a single reusable block suitable for array-level integration, keeping the interconnect between the two stages short to minimize loading on the sensed node.

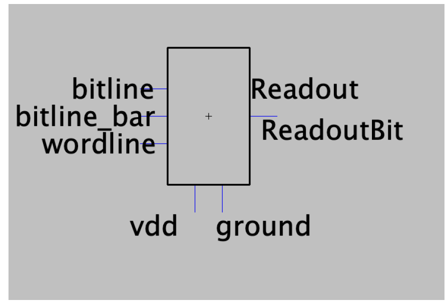

Fig. 12: Hierarchical icon of the complete 1-bit readout circuit.

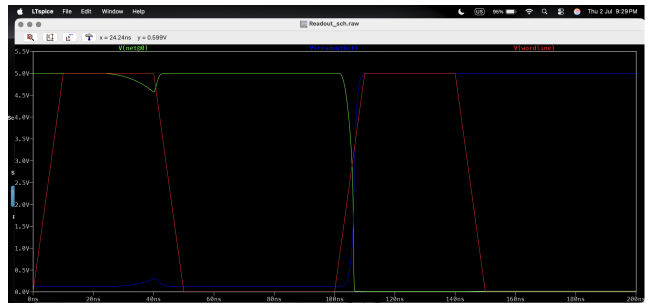

Fig. 13: NGSpice simulation of the proposed 1-bit readout circuit under complementary bitline inputs.

TABLE III: 1-Bit Readout Circuit Simulation Results

BL	BLB	WL	ReadoutBit	Logic
5 V	0 V	5 V	0.043 V	0
0 V	5 V	5 V	4.64 V	1

The circuit was verified under complementary bitline conditions with WL held at 5 V and a 20 fF load capacitor connected to the output node to emulate downstream loading. Fig. 13 shows the resulting simulation waveform.

The results, summarized in Table III, show that when BL = 5 V and BLB = 0 V, the output was approximately 0.043 V, corresponding to logic 0, and when BL = 0 V and BLB = 5 V, the output was approximately 4.64 V, corresponding to logic 1. Both outputs sit close to the supply rails rather than at an intermediate voltage, confirming that the Schmitt-trigger-assisted stage successfully regenerates a valid digital response bit from the SRAM cell's startup state.

E. WL Decoder

A row-selection stage is required to access individual cells inside the 4×4 array without disturbing the remaining, unselected rows. The WL Decoder performs this function: given a row address, it drives the corresponding wordline high while holding all other wordlines low, so that exactly one row of PUF cells is enabled for readout at a time. This addressing scheme allows a single readout path to be time-multiplexed across all sixteen SRAM PUF cells rather than requiring a dedicated readout circuit per cell, which keeps the overall array area compact.

Fig. 14 shows the Electric VLSI schematic of the WL Decoder implemented for the array.

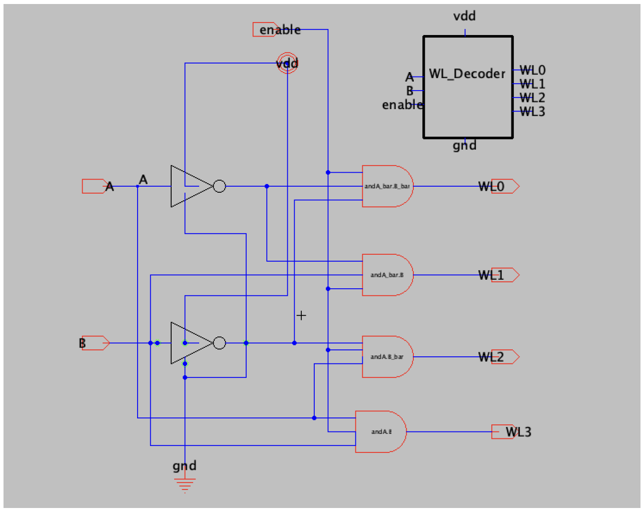

Fig. 14: Electric VLSI schematic of the WL Decoder used for row selection in the PUF array.

F. Bit Detector

The Bit Detector is the final stage of the readout path. It receives the ReadoutBit signal produced by the Schmitt-trigger-assisted readout circuit and evaluates it against valid CMOS logic levels before the bit is used as part of the PUF response. Because the Schmitt Trigger already regenerates a full-swing digital signal, the Bit Detector's role is to interface this signal cleanly with the rest of the digital system, so that each of the sixteen response bits produced during a full array scan is presented as a well-defined logic value rather than a raw analog readout voltage.

G. 4×4 PUF Array

The complete array consists of sixteen 6T SRAM PUF cells arranged in four rows and four columns, each cell contributing one response bit toward the 16-bit PUF output. The cells within a row share a common wordline, while the cells within a column share a common bitline pair, following the conventional SRAM array organization. The WL Decoder selects one row at a time by asserting its wordline, and the bitline pairs of the selected row carry the corresponding startup states to the readout path. The Schmitt-trigger-assisted readout circuit and Bit Detector are reused across the array, sensing the selected cell's state and producing a valid response bit before the decoder moves on to the next row.

Fig. 15 shows the Electric VLSI schematic of the complete 4×4 SRAM PUF array, built hierarchically from the previously described SRAM cell, WL decoder, and readout blocks.

Fig. 16 shows the NGSpice simulation of the array, illustrating the response-bit generation obtained when a given row is selected.

V. Layout Implementation and Verification

The project was implemented using Electric VLSI with a hierarchical design methodology. Each reusable circuit block was created as an independent cell with a schematic, icon, and layout where applicable. This organization improves readability and makes the final array easier to build and debug.

The layout implementation followed standard CMOS full-custom layout practices. PMOS devices were placed in the pull-up network and NMOS devices in the pull-down network. Power and ground rails were routed consistently across hierarchical cells. Contacts, wells, and routing layers were arranged to keep the design compact while maintaining readability.

For the PUF cell and readout circuit, layout matching is important because transistor mismatch determines the SRAM startup response. Therefore, the cross-coupled inverter devices should be placed symmetrically where possible. For the Schmitt Trigger, feedback routing should be kept short and clear to avoid unnecessary parasitic loading.

Design-rule checking (DRC) should be performed in Electric. Table IV provides a place to summarize the layout status for each block.

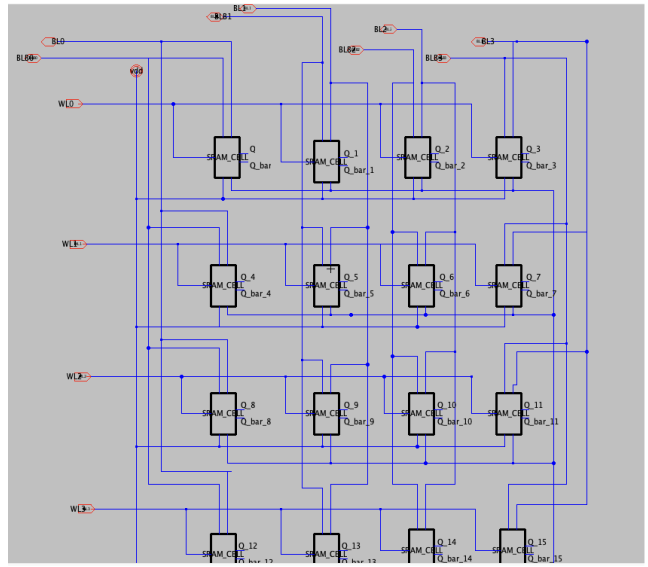

Fig. 15: Electric VLSI schematic of the complete 4×4 SRAM PUF array.

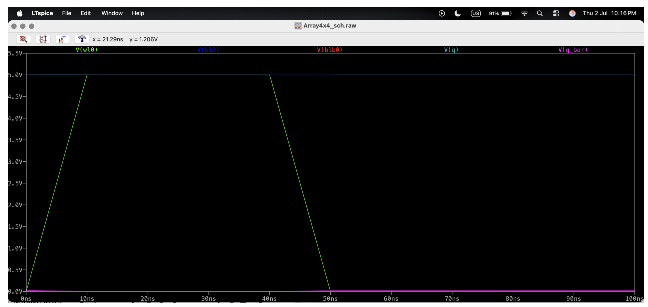

Fig. 16: NGSpice simulation of the 4×4 PUF array showing selected response-bit generation.

VI. Simulation Setup

All circuit simulations were performed using NGSpice on netlists exported from Electric VLSI. Unless otherwise stated, the results are schematic-level simulations. No fabricated-silicon measurements are claimed.

TABLE IV: Layout Verification Summary

Block	Layout	DRC	Notes
Inverter	Done	To verify	Replace note
SRAM Cell	Done	To verify	Replace note
Schmitt Trigger	Done	To verify	Replace note
Readout	Done	To verify	Replace note
WL Decoder	Done	To verify	Replace note
Bit Detector	Done	To verify	Replace note
4×4 Array	Done	To verify	Replace note
Top-Level Design	Done	To verify	Replace note

TABLE V: General Simulation Conditions

Parameter	Value
Tool	NGSpice
Design Tool	Electric VLSI
Supply Voltage	5 V
Schmitt Input Sweep	0–5 V
Sweep Step	0.05 V
Readout Load Capacitor	20 fF
Readout WL	5 V

A. General Simulation Conditions

The main simulation conditions are summarized in Table V.

B. Schmitt Trigger Testbench

The Schmitt Trigger was evaluated using a DC transfer simulation. The input voltage was swept from 0 V to 5 V while recording the output voltage. The purpose of this simulation was to verify logic swing and switching threshold.

C. Readout Testbench

The readout circuit was evaluated under two complementary bitline cases. In both cases, WL was held at 5 V. The first case used BL = 5 V and BLB = 0 V, while the second case used BL = 0 V and BLB = 5 V. The output voltage was compared against valid CMOS logic levels.

D. Array Testbench

The array-level testbench should activate one wordline at a time and observe the corresponding readout response. This verifies that the WL decoder and readout path operate correctly when integrated into the 4×4 PUF array.

VII. Results and Discussion

A. Schmitt Trigger Results

The Schmitt Trigger produced a full output swing under DC sweep simulation. The measured output was approximately 5.00 V when the input was 0 V and approximately 0.043 V when the input was 5 V. The transition occurred around $V_{TH} \approx 2.7$ V. This confirms that the circuit functions as a digital sensing stage and is able to regenerate clean logic levels.

The main advantage of the Schmitt Trigger is its hysteresis. Compared with a conventional inverter, hysteresis reduces sensitivity to small variations around the switching threshold. This is useful for SRAM PUFs because weak cells can produce analog-like or noisy intermediate voltages during startup or readout.

TABLE VI: Baseline SRAM PUF and Proposed Readout Comparison

Feature	Baseline SRAM PUF	Proposed Design
Entropy source	SRAM mismatch	SRAM mismatch
Readout	Direct/inverter based	Schmitt Trigger assisted
Noise immunity	Lower	Higher
Output stability	More sensitive to weak bits	Improved by hysteresis
Area cost	Lower	Slightly higher
PUF suitability	Functional baseline	More reliable readout

B. Readout Results

The 1-bit readout circuit correctly converted complementary bitline conditions into valid digital logic outputs. The result of 0.043 V represents a strong logic 0, while the result of 4.64 V represents a valid logic 1. This confirms that the Schmitt-trigger-assisted readout can sense the SRAM cell state and produce a stable response bit.

C. Baseline Versus Proposed Readout

A baseline SRAM PUF relies on the startup state of the 6T cell. If the internal node is read directly or through a conventional inverter, the output may be more sensitive to weak mismatches and noise. The proposed design adds a Schmitt Trigger after the SRAM cell. This increases transistor count, but improves readout robustness.

D. Reliability Discussion

The proposed Schmitt-trigger-assisted sensing does not eliminate the need for deeper reliability evaluation, but it improves the first stage of response extraction. The next step is to evaluate repeated power-up behavior, mismatch sensitivity, and environmental variation. A complete reliability study should include Monte Carlo simulation, supply-voltage sweep, temperature sweep, response balance, and BER estimation.

The BER metric can be defined as

$$BER = \frac{\#\text{ Bit Errors}}{\#\text{ Bits} \times \#\text{ Evaluations}}. \tag{2}$$

This metric compares repeated PUF responses against a reference response. In this project, the current verified results focus mainly on transistor-level functionality and readout correctness, while full BER evaluation is reserved for future work.

VIII. Limitations and Future Work

The current work verifies the main circuit blocks using schematic-level simulations. Several limitations remain. First, the reported results do not represent fabricated-chip measurements. Second, layout-extracted simulations should be performed to include parasitic effects. Third, Monte Carlo mismatch simulations are required to evaluate the PUF behavior statistically. Fourth, the complete 4×4 array should be tested under repeated power-up conditions to identify unstable bits.

Future work should include:

- Layout-extracted simulation for the SRAM cell, Schmitt Trigger, and readout block.
- Monte Carlo mismatch analysis to evaluate response randomness and stability.
- VDD sweep to examine supply sensitivity.
- Temperature sweep to examine environmental stability.
- Repeated power-up simulation to identify unstable bits.
- Response balance calculation for the 16-bit output.
- Bit error rate estimation using repeated evaluations.

IX. Conclusion

This paper presented a compact Schmitt-trigger-assisted SRAM PUF design for IoT hardware security. The complete architecture, comprising the CMOS inverter, 6T SRAM cell, Schmitt Trigger, readout circuit, WL decoder, Bit Detector, and 4×4 array, was successfully implemented hierarchically in Electric VLSI and verified through transistor-level NGSpice simulations. The SRAM cell provides the process-variation-based entropy source, while the Schmitt Trigger improves readout robustness by introducing hysteresis and rejecting small voltage fluctuations around the switching threshold.

The Schmitt Trigger simulation showed full logic swing, producing approximately 5.00 V for a low input and 0.043 V for a high input, with a switching threshold near 2.7 V. The complete 1-bit readout circuit produced valid digital outputs under complementary bitline conditions: approximately 0.043 V for logic 0 and 4.64 V for logic 1. These results confirm that the Schmitt-trigger-assisted sensing scheme improved the reliability of SRAM startup-state detection and that the readout circuit is suitable for integration into the complete 4×4 PUF array.

Future work will extend this evaluation beyond schematic-level, single-condition simulation. In particular, Monte Carlo analysis is needed to characterize the statistical behavior of the response across process variation, process-voltage-temperature (PVT) corner analysis is needed to assess robustness under realistic operating conditions, the design should be scaled to larger arrays to evaluate area and routing overhead, and the layout should ultimately be carried through fabrication to validate the architecture on silicon.

Acknowledgment

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